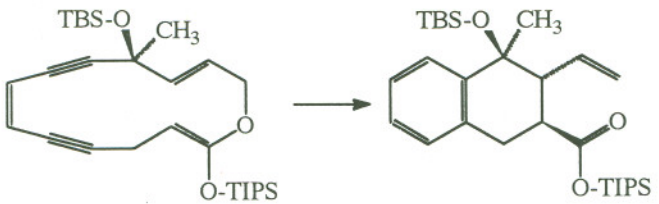
Organic Chemistry Cumulative Examination
April 24, 1993

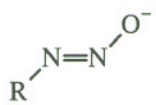
1. The following transformation involves two "named" processes, both of which are quite well-known and important to organic chemists. Name these processes and show what they are by means of the usual "electron pushing" diagrams.



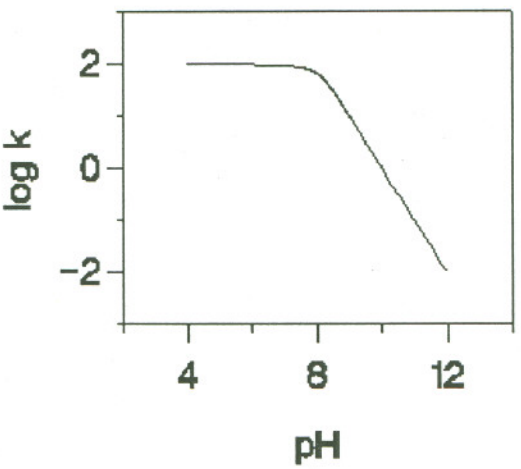
TBS = t-BuMe$_2$Si
TIPS = (i-Pr)$_3$Si

Reference: P.A. Magriotis and K. D. Kim, *J. Am. Chem. Soc.* **115**, 2972 (1993).

2. Alkanediazoates decompose in aqueous solution to ultimately give nitrogen and alcohols and/or alkenes. The kinetics of the decomposition of E-2-butanediazoate was recently studied. The pH dependence of the rate constant, corrected for buffer concentration, is shown below. Deduce a mechanism for the decomposition that is consistent with this data. The pK_a of the diazoic acid is 8.83.

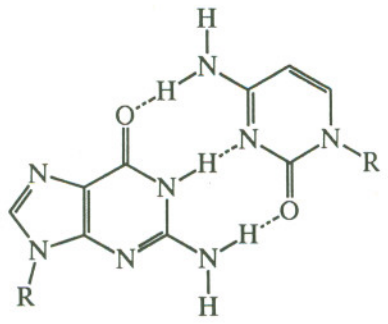


R= 2-butyl

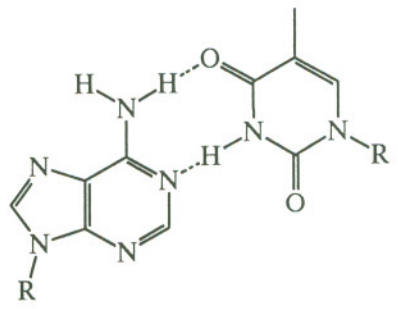


Reference: J. I. Finneman, J. Ho, and J. C. Fishbein, *J. Am. Chem. Soc.* **115**, 3016 (1993).

3. This problem is about nucleic acid structure, so I'll start by reminding you of some background information. Nucleic acids—DNA—are composed of a sugar–phosphate backbone and the heterocyclic bases guanine (G), cytosine (C), adenine (A), and thymine (T). The usual structure is the double stranded helix, where the bases form specific hydrogen–bonded pairs, the specificity being G with C and A with T. The specificity is mostly due to the hydrogen bonding pattern between these base pairs.



G---C



A---T

R = backbone

In the double helix, the sugar–phosphate backbone is on the outside, exposed to solvent, while the base pairs are inside, where they are not solvent-accesible.
Although double-helical DNA the most common naturally occuring structure, there is a wide variety of other structures that nucleic acids can adopt, depending on the nucleic acid sequence and the solution conditions. It is particularly noteworthy to remember that base pairs other than G–C and A–T are possible, as long as the bases have matching patterns of hydrogen bond donors and acceptors.

This problem is about the sequence C-C-C-T-T-T-T-C-C-C. As you can see, there are no A's and G's here, so this sequence cannot form the normal double helical DNA. Under basic conditions (pH > 7.3), the proton NMR spectrum of this sequence shows a broad signal at δ = 11 ppm. The signal is due to the imino proton of T residues, and the broadness is due to exchange with the solvent. When the solution pH is lowered (to 5.5), a new signal is observed in the same region; to be precise, at δ = 16 ppm. Also, the original signal at 11 ppm becomes quite sharp and narrow. Explain these observations.

Reference: D. S. Pilch and R. H. Shafer, *J. Am. Chem. Soc.* **115**, 2565 (1993).

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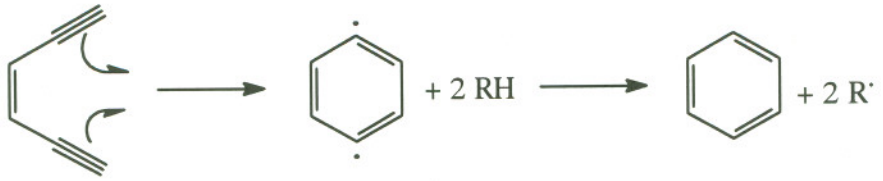
Answer Key

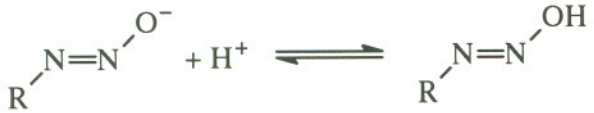
1. Bergman rearrangement:



(RH is obtained from the solvent, or whatever other source is available.)

Claisen rearrangement:

2.



rate determining step

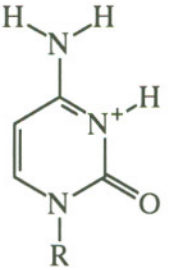
$R-N\overset{+}{\equiv}N \longrightarrow R^+ + N_2$ $\quad$ ($R^+ = CH_3CH_2CH^+CH_3$)

$CH_3CH_2CH^+CH_3 + {}^-OH \longrightarrow CH_3CH_2\overset{OH}{\overset{|}{C}}HCH_3$ and/or $CH_3CH{=}CHCH_3 + H_2O$

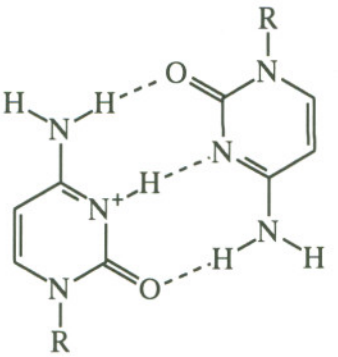
The essential point here is that the rate determining step is the decomposition of the diazoic acid. Below the pKa, the diazoate exists as the protonated form, i.e., the acid. So the rate constant is independent of pH. Above the pKa, the amount of diazoic acid is proportional to the H^+ concentration, so the rate constant is linear in pH.

Note that it is never possible to determine, from the kinetics, what happens after the rate detemining step. The diazoic acid may decompose directly into hydroxide, nitrogen, and the carbocation—the information given in the problem does not preclude this possibility (however, additional information presented in the reference does make it unlikely). However, you should know enough about carbocation chemistry to deduce that a carbocation is probably involved as an intermediate in the formation of the alkene and/or alcohol.

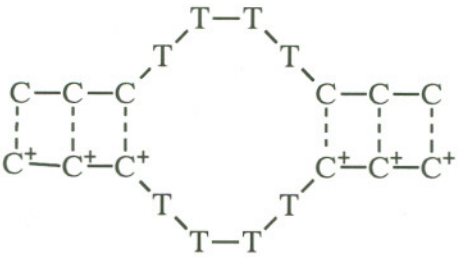
3. At the high pH, the molecule is single stranded. All the bases are exposed to solvent, thus proton exchange is easy, and the imino proton signal is broadened. At the lower pH, half the cytosine residues are protonated. The new protons are also imino protons, so their signal is in the same region as the imino protons of the thymines.



The protonated cytosine residues now have the correct pattern to form hydrogen bonds to unprotonated cytosine residues.



The molecules are now double-stranded. Even though the thymine residues are not hydrogen-bonded into base pairs, this structure still places the bases where they are not solvent-accesible.



This slows down the proton exchange processes that causes the broadening of the signal in the original, single-stranded structure, causing the NMR signal to sharpen and narrow.